

Solar-Powered Irrigation Systems

This article describes the different components of solar-powered irrigation systems, various specifications, different techniques used, the current scenario in India, the challenges faced by the consumers and the future scope of solar-powered irrigation systems

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Solar-powered water pumping systems can find application in town water supply, livestock watering and irrigation. Solar-powered irrigation system is an application of solar-powered water pumping system used in paddy fields, gardens for watering the plants, vegetables, etc. A typical example of solar-powered irrigation system is shown in Fig. 1.

Some advantages of a typical solar irrigation system are:

1. It makes irrigation possible in remote areas
2. Is environment friendly
3. No grid connection required
4. No electricity bills to be paid
5. No fuel required
6. Is durable, requiring minimal maintenance

Essential components

The actual components and hardware requirement depend on the type of irrigation system, such as surface irrigation where water is moved across the surface of agricultural lands, localised irrigation like spray or drip or trickle system (Fig. 2) where water is applied to each plant or adjacent to it, and sprinkler irrigation where water is piped to one or more central locations within the field and distributed by overhead high-pressure sprinklers or guns.

The major components of a typical solar-powered pumping system include a solar panel array that powers a bore-well pump or surface pump. The size of the complete system is dependent on size of the pump, the amount of water required per day and power available from the solar panels. The water is often pumped from a bore-

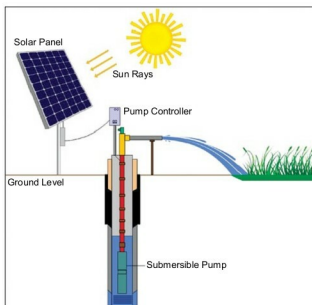


Fig. 1: A typical solar-powered irrigation system (Courtesy: www.taiyosolar.in)

well or stream into a storage tank or directly into the field. The solar-powered irrigation system has three main parts: motor pump, pump controller and solar panel.

Motor pump. The pump has a motor running on electricity generated by the solar panel. Depending on the type of motor (AC or DC), the voltage of the solar pump motor can be AC or DC. DC motors are mostly used for small to medium applications, such as garden fountains, landscaping, drinking-water for livestock, or small irrigation projects. Most DC pumps run at 24 volts rather than 12 volts. However, smaller systems for 12 volts do exist for light-duty applications, while systems running on 48 volts also exist for heavier applications. Power of a motor is specified in either watts or horsepower (HP).

Pump controller. There are two types of pump controllers: inverter and

variable frequency drive (VFD). If an AC solar pump is used, an inverter becomes necessary to change the DC from the solar panels into AC. The supported power range of a typical inverter extends from 0.15kW to 55kW, with the higher power inverter being used for larger irrigation systems.

The solar panel and inverter must be sized accordingly to accommodate the inrush characteristic of

an AC motor. Since AC pump requires high power at the start, the inverter should be able to handle this extra start-up load. Sometimes a VFD controller is used to ensure that the pump motor gets proper voltage and current. Some typical specifications for various AC pumps and pump controllers are listed in Table I.

Many solar DC pumps require a special controller if they are to be powered directly by PV modules (without batteries). The controller or linear current booster (maximum power point tracker) enables the pump to start and run in low light conditions during a cloudy day, or early morning and evening. With a battery power source, the controller may not be required at all for a DC pump.

A special controller may also be employed for converting 12-volt battery power to higher voltage to run the pump at top speed. The DC pumps are